

Artificial Intelligence – Study Notes

Notes for RAG, Embeddings & Vector Database Project

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, understanding language, recognizing images, solving problems, and making decisions.

2. Machine Learning

Machine Learning (ML) is a branch of AI in which computers learn patterns from data instead of being explicitly programmed for every task. Common types include supervised learning, unsupervised learning, and reinforcement learning.

3. Deep Learning

Deep Learning is a type of machine learning that uses neural networks with multiple layers. It is widely used for image recognition, speech processing, natural language processing, and recommendation systems.

4. Natural Language Processing

Natural Language Processing (NLP) enables computers to work with human language. Applications include chatbots, translation, sentiment analysis, text classification, question answering, and summarization.

5. Large Language Models

Large Language Models (LLMs) are AI models trained on large amounts of text. They can generate text, answer questions, summarize information, translate languages, and assist with many language-based tasks.

6. LLM Hallucinations

An LLM hallucination occurs when a language model produces information that sounds convincing but is incorrect, unsupported, or not present in the available information. Retrieval-Augmented Generation can help reduce this problem by providing relevant external context.

7. Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with text generation. Instead of relying only on information stored in an LLM, a RAG system first retrieves relevant information from a document collection or database and then gives that information to the LLM as context for generating an answer.

8. Document Processing

Document processing converts source documents into usable text. For a PDF-based RAG system, the document can be read, its text extracted, cleaned, and prepared for further processing.

9. Text Chunking

Large documents are divided into smaller pieces called chunks. Chunking makes retrieval more efficient because the system can search for the most relevant portions of a document instead of processing the entire document at once. Chunk overlap can help preserve context between neighboring chunks.

10. Embeddings

An embedding is a numerical representation of text. Similar meanings tend to produce vectors that are close to each other in embedding space. Embeddings allow a system to perform semantic similarity search rather than relying only on exact keyword matching.

11. Semantic Similarity

Semantic similarity measures how closely two pieces of text are related in meaning. A question can be converted into an embedding and compared with document-chunk embeddings to find the most relevant information.

12. Vector Databases

A vector database stores and searches numerical vectors efficiently. In a RAG application, document chunks are converted into embeddings and stored in a vector index. When a user asks a question, the query embedding is compared with stored vectors to retrieve relevant chunks.

13. FAISS

FAISS, developed by Meta AI, is a library for efficient similarity search over dense vectors. It can be used as the vector-search component of a small RAG application.

14. RAG Pipeline

A typical document-based RAG pipeline is: Document → Text Extraction → Chunking → Embeddings → Vector Database → Retrieval → LLM → Answer. The retrieved chunks provide context that helps the model answer questions about the source documents.

15. Example

Suppose a student uploads AI notes and asks, 'What is RAG?' The system converts the question into an embedding, searches the vector database for relevant chunks, retrieves the chunks discussing RAG, and supplies them to the language model. The model then generates an answer based on that retrieved context.

16. Advantages of RAG

RAG can connect an LLM to private or updated documents, provide source-grounded context, and make document question-answering systems more useful. Retrieval quality, chunking strategy, embedding model, and the quality of the source documents all affect the final result.

Key Terms: AI, Machine Learning, Deep Learning, NLP, LLM, RAG, Hallucination, Chunking, Embeddings, Semantic Similarity, Vector Database, FAISS, Retrieval.